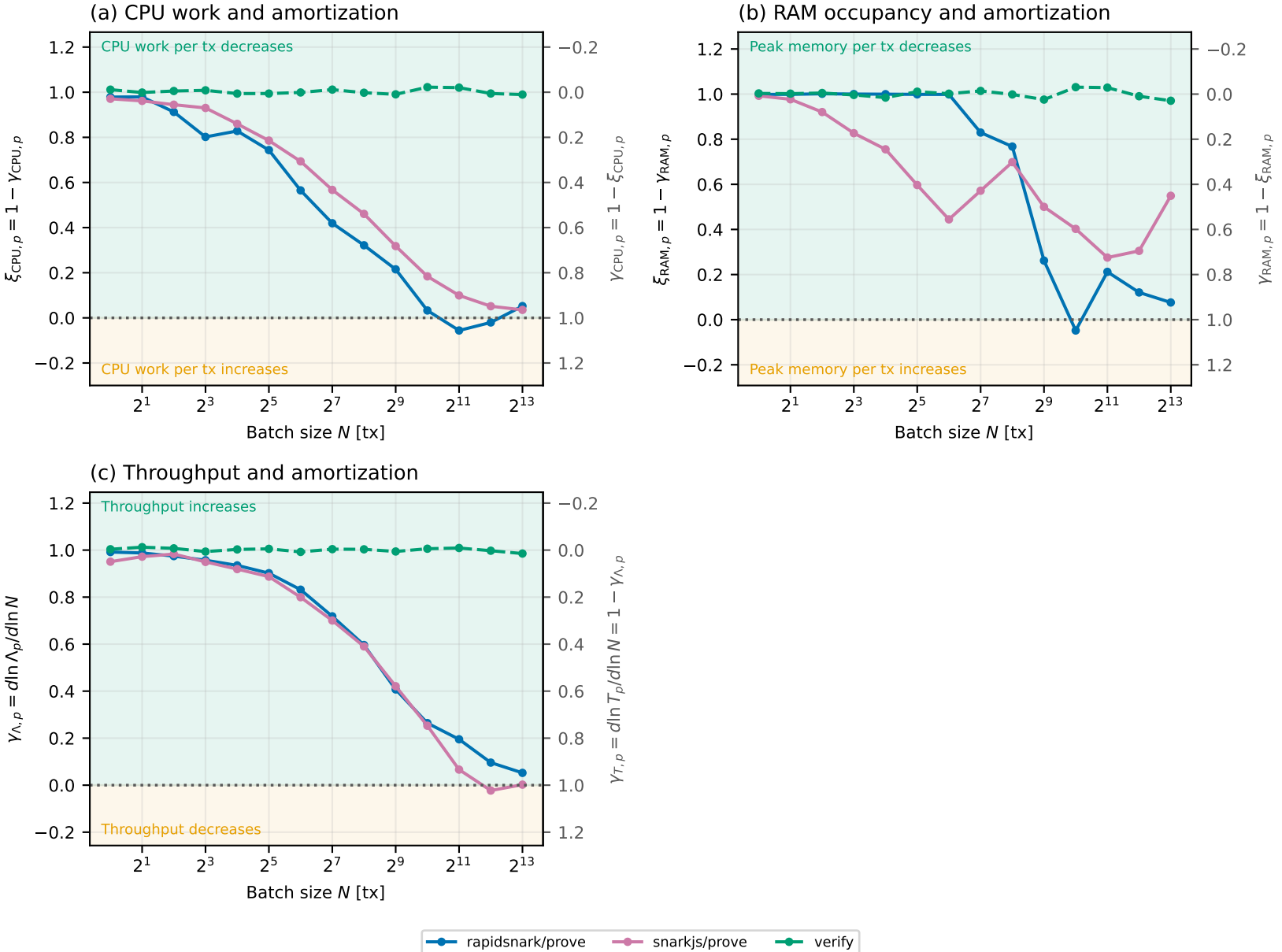
Resource amortization and throughput gain (dimensionless)
Higher on the left axis = greater batching gain; right axes = 1 - left axes
(a) CPU work and amortization
CPU work per tx decreases
CPU work per tx increases
$\xi_{CPU,p} = 1 - \gamma_{CPU,p}$
$\gamma_{CPU,p} = 1 - \xi_{CPU,p}$
Batch size N [tx]
(b) RAM occupancy and amortization
Peak memory per tx decreases
Peak memory per tx increases
$\xi_{RAM,p} = 1 - \gamma_{RAM,p}$
$\gamma_{RAM,p} = 1 - \xi_{RAM,p}$
Batch size N [tx]
(c) Throughput and amortization
Throughput increases
Throughput decreases
$\gamma_{\Lambda,p} = d\ln\Lambda_p/d\ln N$
$\gamma_{T,p} = d\ln T_p/d\ln N = 1 - \gamma_{\Lambda,p}$
Batch size N [tx]
rapidsnark/prove
snarkjs/prove
verify